

NUMERICAL ANALYSIS

PROBLEM SET 5

ROB TOMPKINS

1. PROBLEM 1

Derive the trapezoid integration scheme as an Adams-Moulton method with $m = 1$ step:

$$\frac{w_{j+1} - w_j}{h} = \frac{1}{2}[f(t_{j+1}, w_{j+1}) + f(t_j, w_j)]$$

to approximate the initial value problem $y' = f(t, y)$ for $a \leq t \leq b$ with initial condition $y(a) = \alpha$. Also derive the associated error term. Let $w_j \approx y_j := y(t_j)$.

- (a) Use the Lagrange form of the polynomial interpolant to write

$$f(t, y(t)) = P_1(t) + R_1(t).$$

Integrate $P_1(t) = L_{1,0}(t)f(t_{j+1}, y_{j+1}) + L_{1,1}(t)f(t_j, y_j)$ from $t = t_j$ to $t = t_j + 1$.

- (b) Set $w_{j+1} = w_j + \int_{t_j}^{t_{j+1}} f(t, y(t))dt$, approximating the 2nd term with $\int_{t_j}^{t_{j+1}} P_1(t)dt$, and derive the method above.
- (c) The error term $R_1(t) = \frac{f''(\xi, y(\xi))}{2}(t - t_{j+1})(t - t_j)$ for f (some $\xi \in (t_j, t_{j+1})$) can be integrated from $t = t_j$ to $t = t_{j+1}$ to get the error associated with the Adams-Moulton scheme:

$$\text{Error} = Mh^3 f''(\mu, y(\mu)).$$

Do this integration to find value of M .

- (d) What is the order of this one-step implicit method?

Solution.

□